

The way hard drives access data is by moving the head across the surface and then seeking the beginning of the data stream, therefore, higher RPM drives have lower seek times and thus faster data rate. This data rate is also dependent on the interface attached to the drive. PATA drives were way slower than SATA / SCSI drives, this had a lot to do with one interface being serial and the other being parallel. While seeking data to alternating extremities leads to reduced data transfer speeds the normal user does not experience them as equally. So whenever such reduced periods would creep up the copied data would be transferred to the hard drive cache which would hold on to the data till the read/write head would seek to the write place. So having a higher cache also helps but anything above 32 MB is a bit overkill.

Hybrid drives

With SSDs becoming more and more accessible owing to reduced chip costs even hard drives started incorporating the chips into their design. This gave birth to the hybrid drive. There are two ways that this is accomplished. You can either use two separate drives (one SSD and one HDD) to create a hybrid solution where data would be transferred to the SSD quickly and then the drive took its own sweet time to transfer it to the HDD. Also frequently used programs and files were stored on the SSD to improve load times.



The true hybrid drive is when there is just one physical disk onto which a few GBs worth of memory chips are placed. They work in a similar fashion as explained above leading to faster transfer speeds and caching of frequently used applications.

Interface

PATA / IDE

ATA stands for Advanced Technology Attachment, its two variants are named PATA and SATA given their mode of operation (Parallel and Serial). PATA used flat and wide 40 pin connectors and were capable of transferring data at speeds of around 100-133 MB/s. This was the maximum potential,